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Decoupling wastewater-related greenhouse gas emissions and water stress alleviation across 300 cities in China is challenging yet plausible by 2030

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Urban wastewater treatment and reuse infrastructure play a vital role in achieving water sustainability; however, the pathways to realize water–climate synergies in planning such infrastructure are not clear. Here we examine the nexus of urban water stress and greenhouse gas (GHG) emissions resulting from expanding wastewater infrastructures across over 300 cities in China. We find that, despite a total increase of 176% in life-cycle GHG emissions, larger-scale wastewater treatment and reclaimed water reuse have nearly tripled the average amount of urban water stress alleviated between 2006 and 2015. However, with an extensive and integrated application of existing low-carbon technologies for wastewater treatment, sludge disposal and water reuse, it is possible to substantially further decouple water stress mitigation from GHG emissions by 2030. Under the optimized scenario, China can reduce wastewater-related emissions by 27% at the national level, while its eastern and northern cities could reduce emissions by over 40% for every unit of alleviated water stress. This study provides insights into the water–climate nexus and outlines feasible pathways to reduce water stress while mitigating wastewater-related GHG emissions.

Well-functioning wastewater treatment systems are crucial infrastructure for improving water sustainability and accessibility by replenishing water supplies and reusing reclaimed water[1–4]. However, given the increasingly higher standards for water quality and stringent environmental regulations, wastewater treatment and reuse have led to a notable growth in energy use and greenhouse gas (GHG) emissions. In China, rapid urbanization and the increasing demand for freshwater lead to the expansion of urban wastewater infrastructure, which in turn exacerbates climate impacts[5–7]. CH_4 and N_2O emissions from wastewater treatment and sludge disposal accounted for approximately 5% of

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China's total non-CO_2 emissions[8]; this value is expected to increase to over 10% in 2050 (ref. 9). The GHG emissions related to wastewater treatment would be even bigger if the life-cycle contributions from upstream energy use, construction of infrastructure, on-site energy use and treatment processes, and sludge disposal are included[10–13]. Therefore, reducing GHG emissions from wastewater treatment should be given a high priority in fast-urbanizing countries like China, where the amount of wastewater treated is already large and continuously growing. In this avenue, improving the accounting and planning of urban wastewater infrastructure throughout their life cycles, is essential for achieving synergy between two key Sustainable Development Goals (SDGs), that is, Clean Water and Sanitation (SDG 6) and Climate Action (SDG 13) (ref. 14).

Over 300 cities in China experience water stress to varying degrees. This is because the growth rate of water demand during urbanization frequently exceeds the provision of freshwater resources, which are in turn, due to climate change, increasingly under stress[5,15]. Meanwhile, to match the growing need for wastewater treatment plants (WWTPs), the number of urban WWTPs in China has increased from 976 in 2006 to over 6,276 in 2015. By contrast, water reuse in China remains in its infancy, accounting for only ~10% of the total volume of treated wastewater in 2015 (ref. 16). This proportion is lower than that achieved approximately a decade ago by other advanced countries in water management, for example, Germany (20%) and Israel (70%) (ref. 17). To further address the threat of water shortages and to promote water availability and equity, in 2015, the Chinese government initiated a series of water management policies. First, these policies were designated to raise the country's effluent quality standard to match the quality for potable waters, for example, lakes, rivers and reservoirs[18]. Furthermore, they set a path of increasing the water reuse rate in cities through new investments in infrastructure, especially in rapidly growing cities and regions facing severe water stress[19].

While China aims to improve water availability and quality for its population by building more wastewater treatment infrastructure, it also aims to achieve the goal of peak carbon emissions by 2030 and carbon neutrality by 2060. However, the increasing demand for wastewater treatment with higher water quality requirements in China has raised concerns over their potential to exacerbate climate change challenges[20–22]. Recent progress has shown that urban wastewater systems in China have a large potential of achieving carbon peak and neutrality by optimizing their energy- and material-intensive treatment processes[21–24]. Furthermore, plans of recycling and reusing larger amounts of water to alleviate water shortages could contribute towards the equally important challenge of decarbonizing China's urban water system[7,25,26]. However, it is still unclear to what degree China's WWTPs can alleviate water stress at the city scale nationwide and how this is going to change given the increasing pressure of cutting GHG emissions throughout their life cycles.

In this article, we unravel the water–climate nexus from the perspective of the life cycle of wastewater infrastructure. Towards this end, we quantified the life-cycle GHG emissions, that is, direct and indirect GHG emissions, of all urban WWTPs using an input–output-based hybrid life-cycle analysis (IO-LCA) from 2006 to 2015. We then mapped water stress alleviation in over 300 cities across China by matching detailed plant-level treatment and reuse data with city-scale water resource and consumption data. On this basis, we proposed an indicator termed the additional climate impact from water stress alleviation (CIWSA), whereby climate impact is represented by GHG emissions. The CIWSA indicator aims to identify the level of decoupling between GHG emissions and changes of water stress at the city scale. This indicator quantifies the extent to which urban water stress alleviation by WWTPs is coupled with their life-cycle GHG emissions, that is, the magnitude of the life-cycle GHG emissions divided by the change of water stress associated with the WWTPs. The CIWSA indicator reveals the challenge of achieving water–climate synergy at the city level in a straightforward and comparative way. To identify pathways of water–climate synergies by 2030, we examined the potential of enhancing water stress alleviation while mitigating life-cycle GHG emissions under various scenarios considering the diffusion and integration of low-carbon technologies at different stages. Our results reveal a challenging yet still plausible scenario to decouple water stress alleviation from an increase in life-cycle GHG emissions of wastewater infrastructure by 2030. This necessitates a wider implementation of existing low-carbon technologies throughout the life cycles of wastewater infrastructure across China's cities.

Uneven wastewater-related GHG emissions across cities

The mapping of GHG emissions attributable to urban WWTPs in China (Fig. 1) reveal, from a life-cycle perspective, that the GHG emissions (measured by their global warming potential on a 100 year timescale, as CO_2 equivalent (CO_2e) increased by 176% during 2006–2015 (from 41 to 113 Mt CO_2e). This was mainly as result of WWTPs expanding by 543% (from 976 to 6,276 plants) and the amount of wastewater treated rising by 308% (Supplementary Figs. 1 and 2). The total life-cycle GHG emissions increased by 99% during the 2006–2010 period, and by 50% during 2011–2015 (Supplementary Fig. 3a). The upstream plant construction (40%, including raw material extraction) and on-site wastewater treatment (46%) showed a bigger contribution to the total life-cycle GHG emissions of WWTPs than downstream sludge disposal (11%) in 2015 (Supplementary Fig. 3a). In addition, although a much smaller life-cycle GHG emissions was generated at the water reuse stage (with a share of 3.4% in 2015), this amount was double the amount in 2006 owing to the rapid increase of investment in water recycling infrastructure. Despite the increase of volume of GHG emissions, the GHG emissions intensity of urban WWTPs (represented by the GHG emissions per unit of wastewater treated) in China has decreased by 61% on average (with a marked reduction from 3.1 kg CO_2e m^{-3} in 2006 to 1.2 kg CO_2e m^{-3} in 2015) (Supplementary Fig. 3b). The GHG emissions intensity of water reuse has also shown an overall trend of rapid decrease (by 28%) during the same time period. By comparison, the GHG emissions intensity of sludge disposal has decreased slowly from 2006 to 2015 (by 30%). The lower GHG intensities were attributable not only to an increase in the operational load rates of plants that better matched their designed capacities, but also to the adoption of more energy-efficient technologies.

The challenge in reducing WWTP-related life-cycle GHG emissions was quite uneven, that is, it was highly concentrated in fast-developing and densely populated cities (Fig. 1a). For example, Beijing, Shanghai, Shenzhen, Chongqing, Suzhou and Chengdu were among the biggest and fastest-growing emitters of WWTP-related GHG in 2015, accounting for 1.5–3.1 Mt CO_2e—almost double the amount of that in 2006. However, the major contributing factors to emissions varied greatly across cities (Supplementary Fig. 4). For Shenzhen, Suzhou and Chengdu, direct and indirect emissions of CH_4, N_2O and CO_2 from wastewater treatment accounted for a large proportion (41–60% of the total emission in 2015). For Beijing, indirect CO_2 emissions related to the building of water reuse facilities were an important source of its life-cycle GHG emissions (55% in 2015), whereas for Chongqing and Shanghai, CH_4, N_2O and CO_2 emissions related to sludge disposal (23–24%) were the main contributors to their life-cycle GHG emissions. The direct CH_4 and N_2O emissions from on-site wastewater treatment in these cities were 8–15 times greater than emissions from wastewater transport and collection; the range of variation was due to the scale of bioreactions and pipeline leaks[27,28]. Although the CH_4 and N_2O emissions from wastewater transport and collection (Supplementary Fig. 5) was comparatively smaller, these emissions, especially for cities in northern and western China, could also be an important contributor to climate change.

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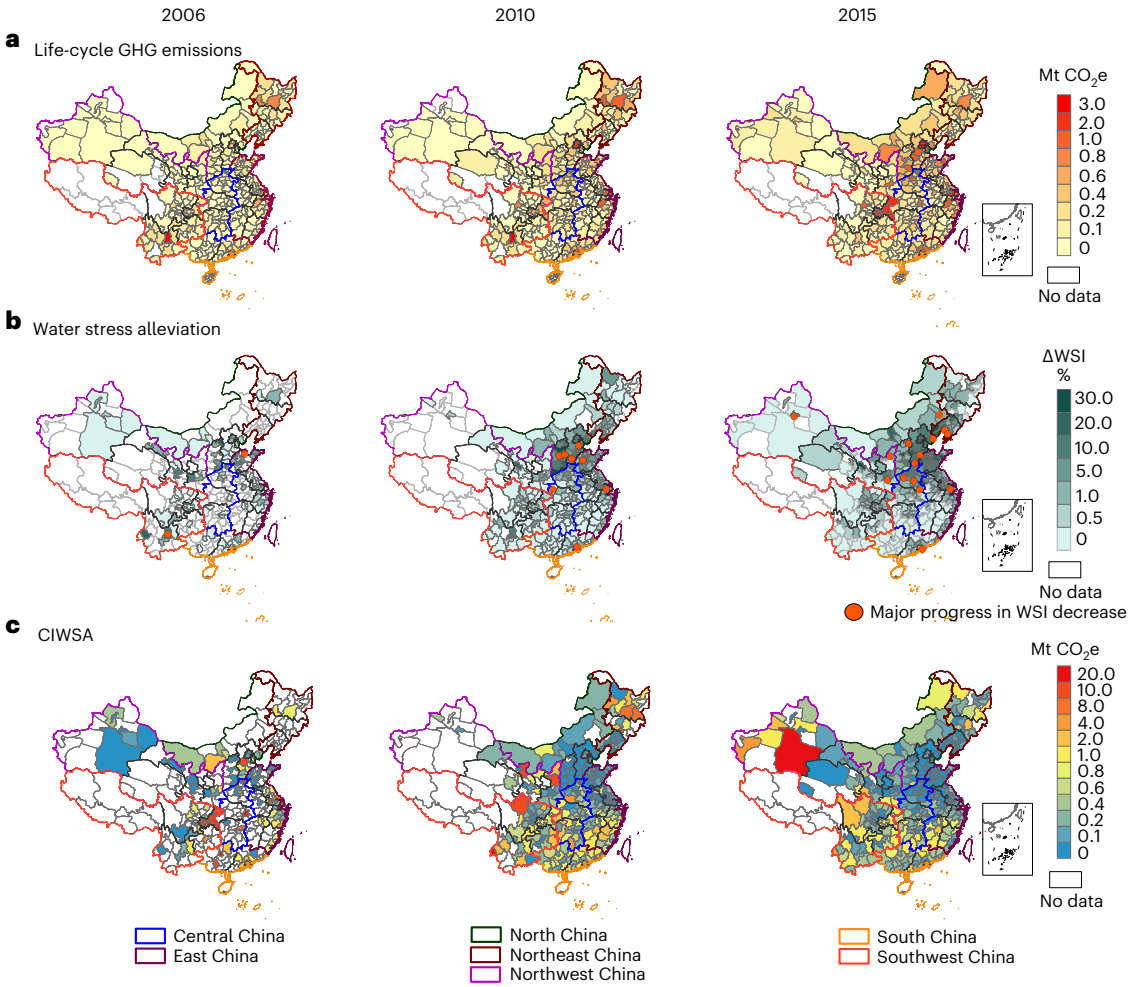

Fig. 1 | Variations in the life-cycle GHG emissions of urban WWTPs and their respective water stress alleviation in 300 cities across China during 2006–2015. a, City-scale life-cycle GHG emissions induced by urban WWTPs in 2006, 2010 and 2015. **b**, City-scale water stress alleviation owing to wastewater treatment, quantified by the difference of WSI with and without current urban WWTPs in 2006, 2010 and 2015 (see details in Supplementary Fig. 7). Life-cycle GHG emissions are represented by CO_2e, which is the sum of direct and indirect CO_2, CH_4 and N_2O emissions weighted by their relative global warming potentials within 100 years (that is, GWP_{100}) and in consideration of climate change feedback[11,21,23]. The GWP_{100} of CH_4 and N_2O is respectively 28 and 265 times that of CO_2 (ref. 70). **c**, Additional life-cycle GHG emissions from water stress alleviation (CIWSA) is defined by the ratio of the life-cycle GHG emissions of WWTPs to the respective variation of water stress in each city. 'Major progress in WSI decrease' indicates the situation that cities (red dots) have downgraded their water stress status (for example, from 'extremely high' to 'high').

Overall, the per-capita life-cycle GHG emissions of urban WWTPs, although with a wide variation among individual cities, was increasing from 2006 to 2015. Medium-size cities (with a population of more than five million) in Guangdong Province, that is, Zhuhai and Zhongshan, Henan Province, for example, Jiaozuo, and Inner Mongolia Region, that is, Alxa League and Ordos, were revealed to maintain a high per-capita WWTP-related life-cycle GHG emissions—all of these cities had over 350 kg CO_2e per capita in 2015 (Supplementary Fig. 6). In comparison, in megacities (with a population of more than 10 million) such as Beijing, Shanghai, Tianjin, Hangzhou and Guangzhou, the per-capita life-cycle GHG emissions were lower, that is, within the range of 81–145 kg CO_2e per capita. This lower per-capita impact can be traced to the advanced treatment technologies and economies of scale of the city's wastewater industries.

The relief of water stress in cities due to WWTPs has increased notably over time, at the national average, that is, from 2.1% in 2006 to 5.7% in 2015 (Fig. 1b). We find the decrease of water stress levels has been mainly contributed by: (1) the enlarged urban wastewater treatment capacity, (2) the increased flow of treated wastewater with high quality that replenishes surface waters and (3) the expanded scale of reclaimed water reuse within cities. The national average of absolute WSI changes increased from 0.01 to 0.10 over 2006–2015, among which Lanzhou, Jiayuguan and Yinchuan in northwestern China all exceeded 1.0, (Supplementary Fig. 7). In 2006, only a few cities in the north, northwest and southwest of China benefited considerably from the water stress alleviation, which was mainly due to the reuse of water from urban WWTPs rather than from wastewater replenishment (Supplementary Fig. 8 and Supplementary Table 1). Since 2010, the relief of water shortage has spread much more widely among other cities. For the two arid cities of Tangshan and Beijing (Supplementary Table 2), by increasing water reuse for landscape, industrial and municipal applications, local water stress was reduced by respectively 12% and 18%. Wastewater replenishment has also played an important role in alleviating water stress, as more urban WWTPs reached a higher standard of discharge water quality. For instance, due to the considerable amounts of reclaimed water and treated wastewater that replenished Taihu Lake water stress in the two nearby eastern cities, namely Suzhou and Wuxi, decreased respectively by 20% and 19%. In 2015, more than 15 cities, mostly located in northern and western China, showed a major progress towards their water stress alleviation (Supplementary

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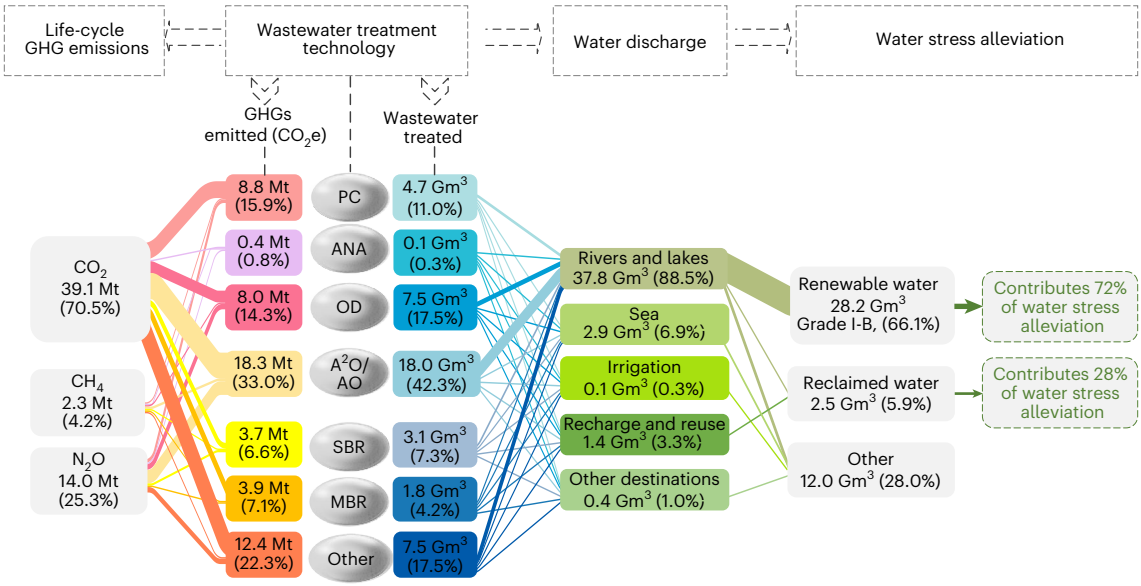

Fig. 2 | Sankey diagram revealing the nexus of water stress alleviation and life-cycle GHG emissions mediated by wastewater treatment technologies in China in 2015. The width of the connecting lines between the boxes is proportional to the magnitude of treated water or GHG emissions.

Table 3). As a result, two northwestern cities, Urumchi and Datong, downgraded their water stress status from 'extremely high' to 'high'[29].

As revealed by the CIWSA indicator, for each unit of water stress alleviated by WWTPs, some of the northwestern cities of China caused the largest life-cycle GHG emissions during 2006–2015 (Fig. 1c). For example, in 2015, Kashgar (CIWSA: 6.7 Mt CO_2e) and Yili (3.7 Mt CO_2e) (both located in Xinjiang Province), had 7–13 times higher life-cycle GHG emissions from water stress alleviation than the national average. This indicates a huge potential in further improving treatment technologies and reusing more water towards achieving water–climate synergy in these cities, although China's northern cities had high total life-cycle GHG emissions; however, due to the larger proportion of reclaimed water as total treated wastewater, in 2015, these cities usually registered a CIWSA value only half as large of cities in the south. This was an important progress as many cities in the north have a much lower water availability and thus a higher reliance on water recycling and reuse.

Influence of technological choices on water–climate nexus

Technology options in wastewater treatment have a profound influence on achieving the synergy of water stress alleviation and life-cycle GHG emissions mitigation (Fig. 2). We investigated the most common WWTP technologies adopted in China, which include physicochemical (PC) treatment, anaerobic (ANA) treatment, oxidation ditches (OD), anaerobic-anoxic-oxic/anoxic oxic (A^2O/AO) treatment, sequencing batch reactors (SBRs), microbial membranes reactors (MBRs) and other technologies, for example, constructed wetlands and oxidation ponds.

Our results reveal that the A^2O/AO, PC and OD technologies made the biggest contributions to water stress alleviation in 2015, accounting for 41%, 12% and 18%, by respectively treating 42%, 11% and 18% of the total wastewater volume. These three technologies also contributed the most to water replenishment, that is, A^2O/AO (43%), PC (10%) and OD (18%), and provided the largest proportions of reclaimed water nationally. Overall, 66% of treated wastewater was replenished as renewable water, contributing to 72% of the water stress alleviation. However, reclaimed water accounted for only approximately 6% of the total treated wastewater, implying the possibility of further increases in the future, for example, by reusing treated wastewater discharged into the sea (7% of total discharge) as cooling water for coastal power plants.

From a life-cycle perspective, the greatest GHG emissions was from plants employing A^2O/AO, PC and OD technologies, which accounted respectively for 33%, 16% and 14% of the total emissions. These technologies produced the biggest share of indirect CO_2 emissions (with A^2O/AO, PC and OD technologies accounting for 60% of total emissions). In terms of intensity, A^2O/AO, OD and SBR technologies generally showed the lowest GHG emissions per unit of wastewater treated (between 1.0 and 1.2 kg CO_2e m^{-3}) while meeting the more stringent discharge standards. In addition to the strong impacts of treatment technologies, life-cycle GHG emission intensities of WWTPs are also influenced by differences of on-site operation and management. This calls for distinctive strategies of technological options among cities to achieve the synergy between water stress alleviation and GHG emissions mitigation. For example, due to the large emissions from the operation of wastewater treatment, A^2O/AO and SBR techniques implemented by cities in Shanxi and Inner Mongolia, have one of the highest GHG emission intensities (Supplementary Fig. 9). In this case, instead of upgrading to new and often expensive technologies, by optimizing the operational management of the wastewater treatment itself, wastewater systems can achieve further decarbonization.

Decoupling of GHG emissions and water stress alleviation

China's wastewater sector faces a big challenge of mitigating GHG emissions in the next decade as the volume of wastewater and its reuse continues to increase (Fig. 3). In the business-as-usual (BAU) scenario, that is, if all technologies remain unchanged, to meet the mounting demand of urban wastewater treatment, the total life-cycle GHG emissions of WWTPs would increase from 113 Mt CO_2e in 2015 (the benchmark year) to an average of 147 Mt CO_2e by 2030 (within the range of 131–162 Mt CO_2e, considering the model uncertainties in Supplementary Table 4; Fig. 3a). In the water reuse enhancement (WRE) scenario, the volume of reclaimed water increases by 428% between 2015 and 2030, therefore increasing the total life-cycle GHG emissions to an average of 164 Mt CO_2e (148–180 Mt CO_2e) by 2030. In the WRE scenario, due to the new construction of water reuse facilities and their energy

usage during their operation, the life-cycle GHG emissions would be 12% higher than in the BAU scenario.

Adopting low-carbon technologies throughout the life-cycle stage of urban WWTP, that is, in wastewater treatment, sludge disposal and water reuse, can open up opportunities for the decoupling of future life-cycle GHG emissions and water stress alleviation. Technological diffusion, as reflected within each of the WRE + LCW (LCW refers to low-carbon wastewater treatment technologies), WRE + LCW + LCSAD (LCSAD refers to low-carbon sludge disposal with anaerobic digesters technologies) and WRE + LCW + LCSAD + LCR (LCR refers to low-carbon water reuse technologies) scenarios, could reduce the life-cycle GHG emissions to respectively 154 Mt CO_2e (141–167 Mt CO_2e), 128 Mt CO_2e (117–138 Mt CO_2e) and 120 Mt CO_2e (110–130 Mt CO_2e) by 2030 (Fig. 3a). With the widest integration of low-carbon technologies in all stages, that is, shown in the WRE + LCW + LCSAD + LCR scenario, the life-cycle GHG emissions would reduce up to 27% of that in the WRE scenario by 2030. Furthermore, within the above scenario, a decrease of emissions would most likely occur between 2025 and 2030—although the total amount in 2030 would still be greater than that of 2015. Among all stages, by 2030, implementing new technologies in the sludge disposal stage has the highest potential for life-cycle GHG emissions reduction (62%), followed by wastewater treatment (22%) and, finally, water reuse (17%).

Another major challenge that needs to be addressed is the increasingly uneven distribution of the future life-cycle GHG emissions of WWTPs among cities (Fig. 3b). Cities in the more developed region of eastern China, due to their rapidly growing need for wastewater and sludge treatment during urbanization, would be most accountable for the total WWTP-related life-cycle GHG emissions. Even under the scenario with the widest integration of low-carbon technologies, that is, the WRE + LCW + LCSAD + LCR scenario, urban WWTPs in eastern China would emit 29 Mt CO_2e (27–32 Mt CO_2e) in 2030, reducing 24% of total emissions, while northern, central and southwestern China would reduce respectively 20%, 17% and 11% of total emissions. Under this scenario, by 2030, the life-cycle GHG emissions of WWTPs in southern and eastern China would reduce respectively 17% and 8% of that in 2015, while in other regions, the life-cycle GHG emissions would be 3–36% higher than that in 2015.

Overall, with the widest integration of low-carbon technologies in all stages (that is, the WRE + LCW + LCSAD + LCR scenario), cities could reduce life-cycle GHG emissions by an average of 26% by 2030, compared with that of the WRE scenario (Supplementary Fig. 10a). It is also possible to have higher reductions (29–32%) for cities in China's northwest and northeast region, as well as megacities such as Shanghai, Chongqing and Tianjin (with a reduction by respectively 34%, 33% and 31%). Cities in eastern and southern China are also expected to lower their GHG emissions related to WWTPs by 2030 from 2015 levels (Supplementary Fig. 10b). Nearly half of the cities in these regions would reduce emissions by 1–33%, with the widest integration of low-carbon technologies between 2015 and 2030. For example, cities like Huizhou, Maoming and Zhongshan would lower their emissions by over 25%.

At the same time, the life-cycle GHG emissions per capita could be reduced by 2–49% across all cities under the widest integrated scenario in comparison with the WRE scenario (Supplementary Fig. 11). In megacities the life-cycle GHG emissions per capita is expected to be 20–34% lower, while in medium and small cities, for example, Chaoyang, Lanzhou and Liuzhou, the life-cycle GHG emissions per capita showed a larger reduction potential (>40%). Despite a rapid increase of water reuse infrastructure, given the low-carbon technologies applied in various stages from plant construction to sludge disposal, these reductions can be achieved.

Both changes in absolute WSI values and levels of water stress show that the future increase of reused water will unleash the potential of water stress alleviation. In this avenue, the national average of absolute WSI changes would be 0.26 under the WRE scenario (with enhanced water reuse) by 2030, double the amount in 2020 (Supplementary Fig. 13). Under the WRE scenario, the degree of water stress alleviation would further increase to 12.0% at the national level by 2030 compared with 9.1% under the BAU scenario (Fig. 4a and Supplementary Fig. 14). In 2030, 20 cities would experience a major decrease of their WSIs under the BAU scenario, thus moving those cities towards a lower-water-stress status. Furthermore, the number of cities registering a major progress in water stress alleviation would increase to 30 under the WRE scenario. The cities registering the largest progress in mitigating their water stress were in the northeast, that is, Panjin and Jinzhou, which would respectively decrease their WSI from 0.8 to 0.3 and from 0.5 to 0.3 and, therefore, improve their status from high-water-stress cities to a less severe status (Supplementary Tables 5 and 6).

Achieving water–climate synergy in the planning of urban WWTPs, however, would not be an easy task. Although, nationally, the life-cycle GHG emissions per unit of water stress alleviated (CIWSA) in 2030 would be lower than 2015 under all future scenarios, stronger decoupling effects of life-cycle GHG emissions and water stress alleviation are observed only in scenarios where low-carbon technologies would be sufficiently adopted (Fig. 4b and Supplementary Fig. 15). Under the WRE scenario in 2030, the average CIWSA of all cities would be ~0.15 Mt CO_2e, which is slightly lower than the BAU scenario (~0.23 Mt CO_2e) and only half of the level in 2015. Under the scenario with the widest integration of technologies, that is, the WRE + LCW + LCSAD + LCR scenario, the average CIWSA of all cities would dramatically drop to ~0.11 Mt CO_2e (Supplementary Fig. 16). For example, much stronger decoupling would occur in megacities such as Shanghai, Beijing and Xi'an under the WRE + LCW + LCSAD + LCR scenario, with the CIWSA ranging from 0.08 to 0.13 Mt CO_2e (33–55% lower than in 2015). By comparison, under the WRE scenario, the decoupling in these cities would be weaker, thus indicating that larger GHG emissions accompany the increase of water stress alleviation. Northeastern cities, with an average CIWSA of 0.02 Mt CO_2e, show the largest advancement in decoupling in 2030, whereby, compared with that of 2015, their CISWA level decreases by ~85%. This is followed by eastern and northern cities with an average CIWSA of respectively 0.05 and 0.08 Mt CO_2e and respectively 41–56% lower than 2015 levels. Northwestern cities, with a CIWSA 2.3 times higher than the national average in 2030, would continue to face the biggest challenge of reducing life-cycle GHG emissions while alleviating water stress. This calls for further efforts in improving technologies at all stages and also enlarging water reuse so that a stronger decoupling can be achieved.

Achieving water–climate synergy in wastewater management

With the threat of future increasing water shortages and climate change, wastewater treatment and reuse will play a more critical role in the sustainable provisioning of water in cities, especially in those already suffering from severe water deficiency[30–32]. In China, the range

Fig. 3 | Future life-cycle GHG emissions of urban WWTPs in China under different scenarios. a, Life-cycle GHG emissions of plant construction, wastewater treatment, sludge disposal and water reuse under different scenarios in different regions of China, whereby the data are presented as mean values ± 1.96 standard errors for a 95% confidence interval. The *n* values represent the number of cities in each region used for accounting. The pie charts illustrate the life-cycle GHG emissions reduction from the adoption of low-carbon technologies at the wastewater treatment, sludge disposal and water reuse stages. **b**, Life-cycle GHG emissions of WWTPs in 2030 under the WRE scenario and the WRE + LCW + LCSAD + LCR scenario (see life-cycle GHG emissions of WWTPs in 2020 and 2025 under the BAU scenario, the WRE scenario and the WRE + LCW + LCSAD + LCR scenario in Supplementary Fig. 12).

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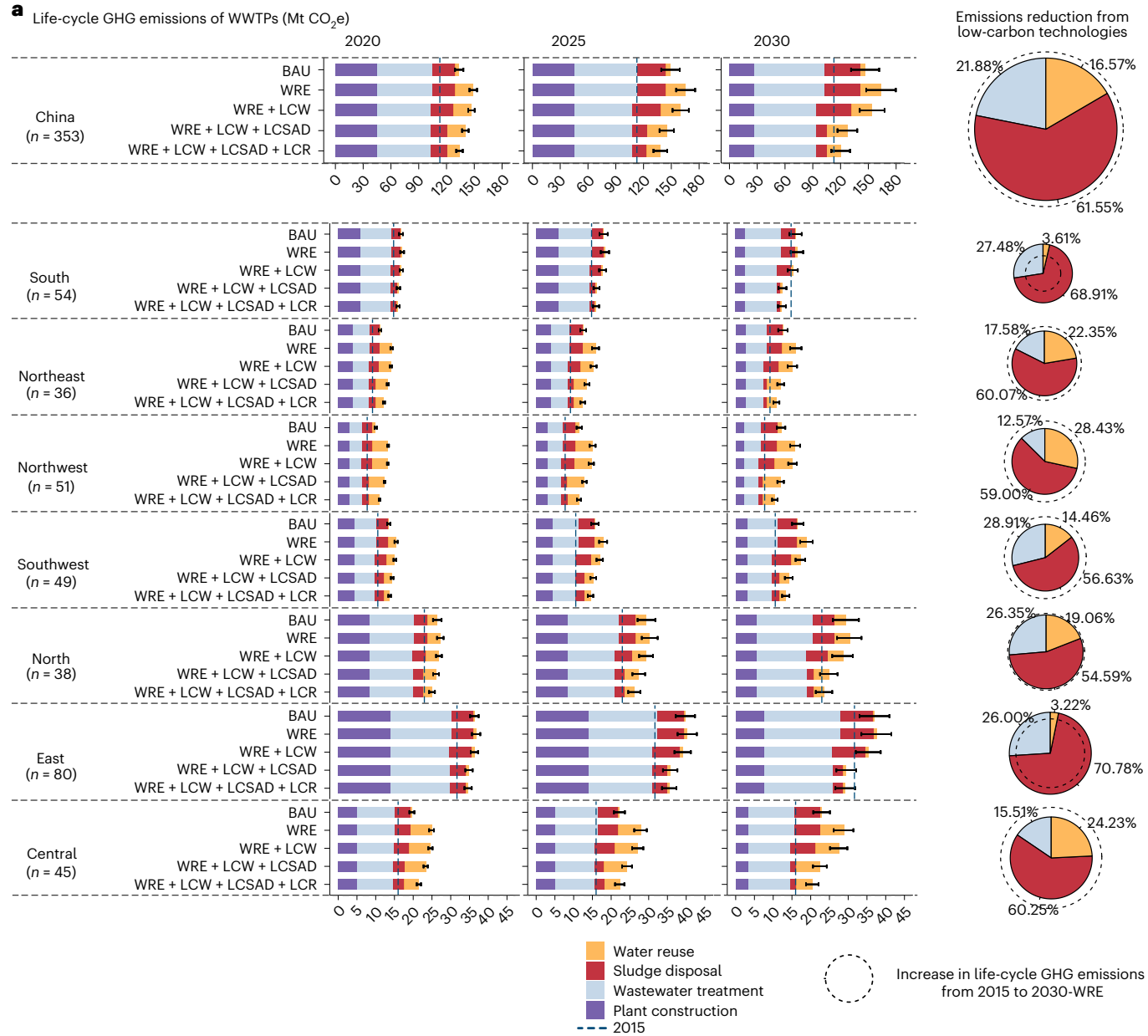
of urban water availability is large and often mismatched with the country's uneven regional economic and demographic growth—in the context of climate change this mismatch is widening even further. While wastewater treatment systems can help relieve water stress in the coming decades, their operation, as well as their upstream and downstream processes and supporting infrastructures, accentuate



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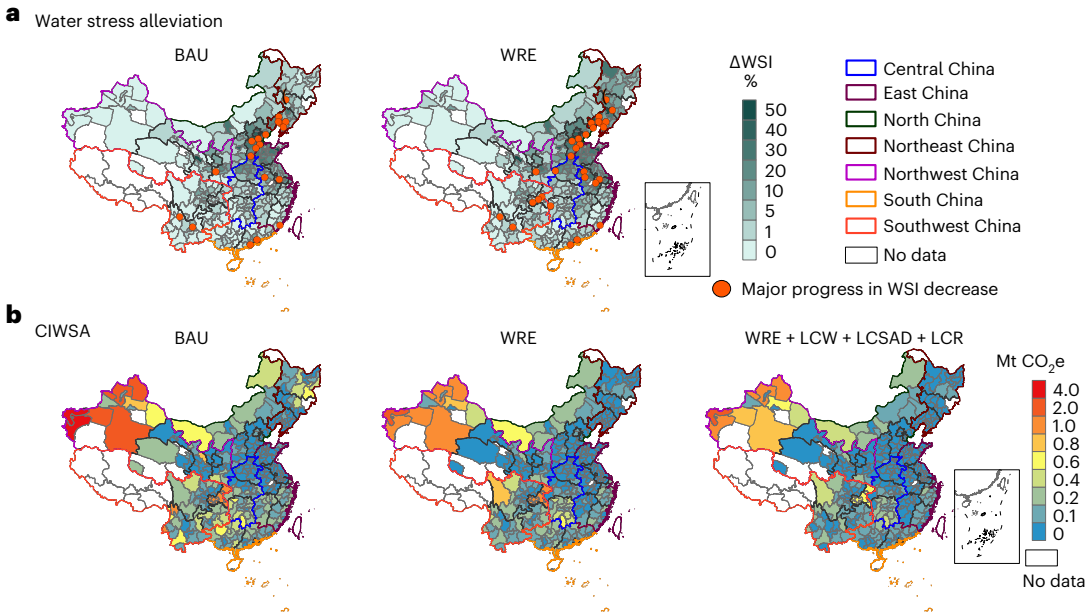

Fig. 4 | Water stress alleviation and additional life-cycle GHG emissions from water stress alleviation (CIWSA) across China's cities in 2030 from WWTPs under different scenarios (see CIWSA in 2020 and 2025 under different scenarios in Supplementary Fig. 15). a, Future water stress alleviation across China's cities. **b**, Future CIWSA under different scenarios. Red dots represent cities exhibiting a major progress in decreasing their WSI.

climate change[6,21,22]. Under China's carbon peak and neutrality goals, decoupling the alleviation of water stress from GHG emissions of the wastewater treatment and water reuse industries is a key challenge that needs to be addressed.

Optimizing available technologies adopted by the wastewater treatment systems, would lead to important benefits in achieving water–climate synergies in Chinese cities. Previously, most technical and management solutions of WWTPs were designed with the intention of reducing electricity use while increasing their water treatment rates and effluent quality[33–35]. Recent discussion, however, has been increasingly concerned with how the supporting technologies and infrastructures may contribute towards achieving carbon peak and neutrality targets for the whole treatment systems[21,22,36]. Our results reveal that the structural optimization of existing treatment technologies alone could realize a ~27% reduction of GHG emissions. Both arid northwest cities and highly populated southwest and central cities have urgent needs for reducing the life-cycle GHG emissions for the sake of alleviating water stress. To address China's ambitious climate goals, future investments in municipal treatment and water reuse projects should provide more carbon-efficient treatment technologies[21,37,38].

For China and other fast-developing regions, decarbonizing WWTPs will be more technologically realistic and economically affordable in the coming decades. This includes, firstly, the potential of methane harvesting, that is, technologies that efficiently capture CH_4 and heat from treatment processes and reuse them to offset emissions[24,39,40]. Currently, methane harvesting from sludge anaerobic digestion by combined heat and power systems is a widely discussed technique of resource recovery[39,41,42]. Although there have been plants that have approached a high self-sufficiency in energy[22,43,44], their GHG emissions, especially from a life-cycle perspective, should be further tested. Secondly, given the high carbon intensities revealed by our study, sludge disposal should also be targeted. Disposal options such as land use, incineration and reuse as building materials are much less carbon intensive than landfills from a life-cycle perspective[45]. Lowering the cost of these technologies would lead to their wider application—currently they are 192–275% more expensive than landfills, the conventional disposal approach in China[46]. Establishing methods and tools to simultaneously monitor GHG emission mitigation and water conservation would be a key step for prioritizing technologies of waste energy and resources recovery. This is critical for lowering GHG emissions per unit of water stress alleviated (CIWSA) at both plant and urban scales.

The unequal distribution of reclaimed water reuse across cities also has strong policy implications for decarbonizing wastewater systems. To meet the growing need for water and wastewater treatment during rapid urbanization, opportunities for improving water efficiency should be combined with enhancing the downstream capability of water recycling[47–49]. We find that the water-deficient cities in the north and northwest had large life-cycle GHG emissions in reusing more treated water from their WWTPs than other regions. However, these cities also had the lowest CIWSA value, that is, had the least impact on climate change for each unit of water stress alleviation, and showed great progress in decreasing water stress levels. Given the applications of more low-carbon technologies by 2030, water reuse in southern cities will also have less GHG emissions. The water reuse rate has been limited in China, that is, <10% on average, much lower than the United Arab Emirates, Israel and Cyprus, reaching respectively 59%, 87% and 97% (refs. 50,51). Although not all Chinese cities could (or should) match the rates achieved by these countries, for those that are heavily reliant on distant and/or alternative sources of water, further development of their water reuse industries is vital for reducing the gap between the urban water demand and water availability in an increasingly changing climate. This will potentially reduce the energy consumption and related GHG emissions in long-distance water transportation projects[52]. However, the additional life-cycle GHG emissions of fast-developing water reuse infrastructure may have to be partially offset by the decarbonization achieved in other processes of the wastewater systems, for example, wastewater treatment and sludge disposal.

More investment in advancing technologies of wastewater treatment and water reuse is critical for the synergistic mitigation of water stress and GHG emissions. To address the issue of water scarcity, the proportion of investment in water reuse in the whole wastewater system has grown by over ten times in China over 2006–2015 (Supplementary Fig. 17a). A major challenge for further increasing investments in

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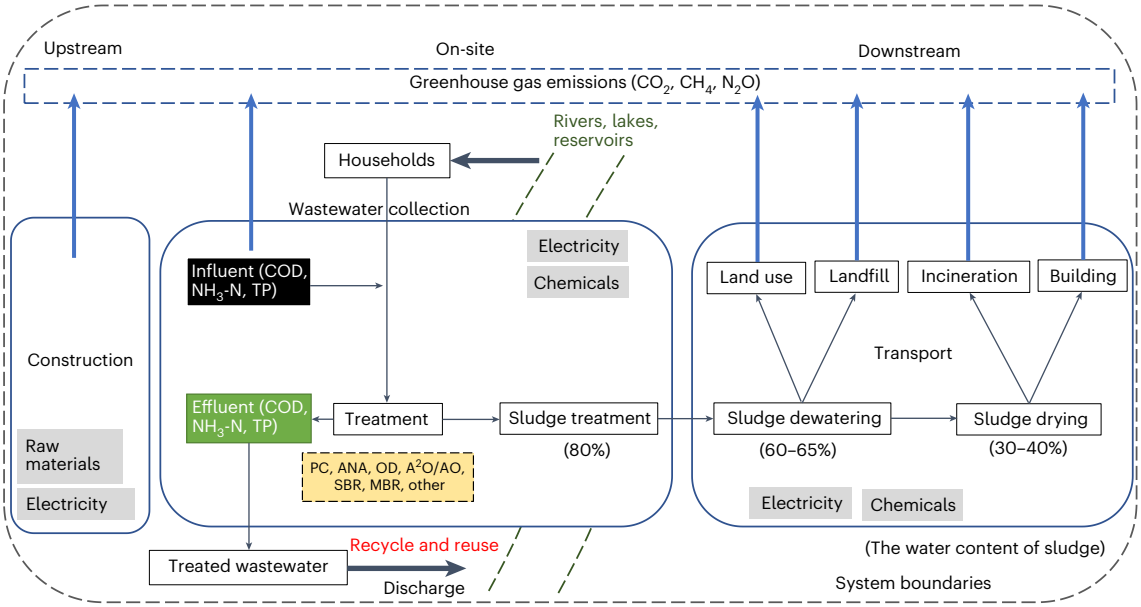

Fig. 5 | Analytical framework for water stress alleviation and GHG emissions related to urban WWTPs from a life-cycle perspective. Water quality indicators for effluent: COD, chemical oxygen demand; NH_3-N, ammonia nitrogen; TP, total phosphorus.

this area is that the current price of wastewater treatment and water reclamation in China is notably lower than those in the developed nations and therefore not attractive to additional private and public investors (Supplementary Fig. 17b). Lower profits could reduce incentives for developing and adopting water-conserving and low-carbon technologies in the wastewater treatment and reuse companies. Higher returns on investments in these projects will be needed and would consequently lead to more low-carbon treatment and water reuse technologies to become affordable to cities at different development stages[37]. Our study reveals key strategies and possible pathways for integrating available and affordable technologies that simultaneously reduce water stress and the GHG emissions of cities.

Methods

Analytical framework and system boundaries

We quantified and tracked the city-scale water stress alleviation and life-cycle GHG emissions induced by urban WWTPs over 2006–2030 using a consistent analytical framework (Fig. 5).

To assess the alleviation effect of urban water stress, the framework included two key factors related to the operation of WWTPs that affects local water availabilities: (1) the treated wastewater discharged as replenishment to natural water bodies, and (2) the reuse of reclaimed water after treatment in plants. The contributions of both factors to the change of urban water stress over time were quantified. On the other hand, to capture the life-cycle GHG emissions of WWTPs, we included life-cycle stages such as infrastructure construction (including material extraction), plant operation (treatment processes) and maintenance, water recycling and reuse, and sludge disposal stages that are directly linked to the performance of WWTPs and the water–climate nexus. Recent studies accounted for GHG emission amounts and intensities of different treatment technologies in China, concerning on-site processes, electricity use and chemical inputs[21,23]. In comparison, here we tracked and quantified GHG emissions from upstream, on-site and downstream processes, including life-cycle impacts of not only wastewater infrastructure but also water reuse infrastructure that are needed to mitigate water stress in cities. In addition, there have been studies reporting CH_4 emissions and the energy requirements related to sewage pipelines[27,28,53]. Therefore, we have also included an estimation of GHG emissions from the collection and transport of wastewater for cities based on an empirical approach widely used in the literature[11,54] (see detailed methods in Supplementary Note 1).

Here we accounted for three types of GHG emissions, that is, CO_2, CH_4 and N_2O, which were identified as the main contributors to WWTP-related life-cycle GHG emissions[11,21,23]. In this system boundary, we accounted for the CH_4 and N_2O emitted from on-site treatment processes and the downstream sludge disposal, as well as the CO_2 emissions driven by energy use and resource inputs in all stages. The analytical processes in the three main life-cycle stages were as follows:

(1) WWTP infrastructure construction. This included GHG emissions embodied in the upstream extraction of raw materials, and the supply of fuels/electricity and other products used for WWTP construction. To reflect the change of construction technologies over time, we assessed the construction-related GHG emissions of the WWTPs based on the year when the facilities were built. According to historical records, approximately 10% of urban WWTPs in China were built during 2000–2003, 20% during 2004–2006 and the remaining 70% during 2007–2010 (ref. 55). On the basis of the average of previous estimates[56,57], the life span of the infrastructure of Chinese WWTPs is generally assumed to be 20 years. Additionally, an increasing number of WWTPs were still under construction during 2011–2015. To show the change of construction technologies over time, we assessed the infrastructure-related GHG emissions of WWTPs based on the year of their construction. To reflect the above considerations, the indirect CO_2 emissions caused by WWTP-related infrastructure were distributed evenly over their 20 year life cycle starting from the year of their construction. This can better match the infrastructure-related emission with the annual accounts at other life-cycle stages of the WWTP, that is, wastewater treatment, water reuse and sludge disposal.

(2) On-site wastewater treatment. Direct release of CH_4 and N_2O from on-site wastewater treatment using various technologies was tracked and quantified. Generally, CH_4 is emitted during aerobic and anaerobic fermentation, while N_2O is released in nitrification/de-nitrification processes (detailed analytical methods for these emissions are provided in Supplementary

Note 1). Indirect CO_2 emissions related to the resources and materials consumed in the operation and maintenance of WWTP systems were also included in this stage. As only a very limited number of WWTPs in China have built-in CH_4 recovery equipment and even a smaller number of those have monitoring data[38,40], recovery of CH_4 as an energy source was not considered. Additionally, the discharge of treated wastewater (meeting certain quality standards) as replenishment in the urban water cycle was analysed for its contribution to urban water stress alleviation.

(3) Water reuse and sludge disposal. The direct and indirect GHG emissions related to the energy (including electricity use) and resource inputs to water reuse and sludge disposal were included in this stage. Given the lack of separate detailed records on water reuse infrastructure, we used investment data obtained from the China Urban Construction Statistical Yearbook[58] to compute the indirect emissions related to water reuse. Statistically, the four types of sludge disposal approaches applied most in China are land use, landfill, incineration and reuse as building materials[39,43]. Direct GHG emissions from sludge disposal mainly include CH_4 emissions from landfill and N_2O emissions from land use[59,60]. To reveal how the water reuse industry could influence the water–climate nexus empowered by WWTPs, we also quantified the contribution of the reclaimed water reused to water stress alleviation across cities.

Accounting for life-cycle GHG emissions of urban WWTPs

We used IO-LCA to compute the life-cycle GHG emissions of WWTPs over different life-cycle stages. Compared with process LCA, the IO-LCA has the advantages of covering the full supply chains[11,61] related to wastewater infrastructure and of acquiring region-specific and time-sensitive emission factors in different life-cycle stages[62,63]. This makes the IO-LCA more suitable for the analysis of time-series life-cycle GHG emissions of WWTPs at the city scale, where energy and material input data of treatment plants are rare. Previous studies have developed various wastewater IO models for life-cycle environmental impact assessments[11,64].

Here we developed a wastewater multi-regional input–output model (W-MRIO) that captured intermediate flows between urban wastewater systems and other socioeconomic sectors (Supplementary Table 7). The model separated the sectors related to wastewater treatment systems, that is, construction of WWTPs, on-site wastewater treatment and water reuse, and sludge disposal, from urban economies on both supply and demand sides. China's MRIO tables for different years were used to reflect the technical change of production over time. The W-MRIO model also distinguished seven types of treatment technologies used in wastewater treatment and reuse and four approaches for sludge disposal while evaluating their GHG emissions. Accordingly, the W-MRIO table included four sections: (I) economic sectors providing goods and services, (II) wastewater-related construction, (III) wastewater treatment and reuse, and (IV) sludge disposal.

The direct GHG emission intensities of the sectors in all four sections were calculated as follows:

$$\boldsymbol{q}_{(n+3)\times 1} = \boldsymbol{Q}_{(n+3)\times 1}\hat{X}^{-1}_{(n+3)\times 1} \quad (1)$$

where $\boldsymbol{q}_{(n+3)\times 1}$ is the GHG emissions intensity vector for economic sector (n), plant construction, wastewater treatment and reuse, and sludge disposal; $\boldsymbol{Q}_{(n+3)\times 1}$ represents the sum of GHG emissions from the respective sectors; and $\boldsymbol{X}_{(n+3)\times 1}$ is the vector of total economic outputs of sectors.

Building on the developed W-MRIO model, the life-cycle GHG emissions (CI) of urban WWTPs and the contributions of different sections were calculated as follows:

$$\mathrm{CI} = \begin{pmatrix} q_{\mathrm{I}} \\ q_{\mathrm{II}} \\ q_{\mathrm{III}} \\ q_{\mathrm{IV}} \end{pmatrix} \left(I - \begin{pmatrix} A_{\mathrm{I,I}} & A_{\mathrm{I,II}} & A_{\mathrm{I,III}} & A_{\mathrm{I,IV}} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & A_{\mathrm{IV,III}} & 0 \end{pmatrix} \right)^{-1} \begin{pmatrix} y_{\mathrm{I}} \\ y_{\mathrm{II}} \\ y_{\mathrm{III}} \\ y_{\mathrm{IV}} \end{pmatrix} \quad (2)$$

where q_{I}, q_{II}, q_{III} and q_{IV} are the direct GHG emission intensity vectors for economic sectors (I), plant construction (II), wastewater treatment and reuse (III), and sludge disposal (IV), which are calculated as follows: $q_{\mathrm{I}} = Q_{\mathrm{I}}\hat{X}_{\mathrm{I}}^{-1}$, $q_{\mathrm{II}} = Q_{\mathrm{II}}\hat{X}_{\mathrm{II}}^{-1}$, $q_{\mathrm{III}} = Q_{\mathrm{III}}\hat{X}_{\mathrm{III}}^{-1}$ and $q_{\mathrm{IV}} = Q_{\mathrm{IV}}\hat{X}_{\mathrm{IV}}^{-1}$. Here, I is the identity matrix; $A_{\mathrm{I,I}}$ is the input coefficient matrix, representing the amount of input from economic sector i required to produce one unit of output from economic sector j, which is given by $A_{\mathrm{I,I}} = Z_{\mathrm{I,I}}\hat{X}_{\mathrm{I}}^{-1}$, where $Z_{\mathrm{I,I}}$) refers to the intermediate input flows in the economic system. Similarly, $A_{\mathrm{I,II}}$, $A_{\mathrm{I,III}}$ and $A_{\mathrm{I,IV}}$ are the input coefficient matrixes, indicating the amount of input from economic sector i required to produce one unit of output respectively from construction (II), wastewater treatment and reuse (III), and sludge disposal (IV). Similarly, the input coefficient matrix $A_{\mathrm{IV,III}}$ is defined by $A_{\mathrm{IV,III}} = Z_{\mathrm{IV,III}}\hat{X}_{\mathrm{III}}^{-1}$, where $Z_{\mathrm{IV,III}}$ represents the disposal inputs of sludge produced by different treatment technologies. Finally, y_{I}, y_{II}, y_{III} and y_{IV} refer to the final demand vectors for the sectors in the respective sections (I, II, III and IV) of the W-MRIO that drive the indirect GHG emissions related to WWTPs.

While improving urban water sustainability, it is also essential to articulate whether the climate costs of building wastewater treatment infrastructure decreases or increases through time. In this study, we proposed an indicator termed the additional climate impact (represented by GHG emissions) from water stress alleviation (CIWSA) to identify the decoupling between GHG emissions and change of water stress at the city scale. CIWSA is calculated as:

$$\mathrm{CIWSA}_i = \frac{\mathrm{CI}_i}{\Delta\mathrm{WSI}_i} \quad (3)$$

where, CI_i refers to the life-cycle GHG emissions of city i attributable to all of its urban WWTPs and $\Delta\mathrm{WSI}_i$ represents the change of urban water stress in city i owing to the treated wastewater replenishment and water reuse from its urban WWTPs (see detailed methods in Supplementary Note 2).

Measurement of variations in urban water stress

The degree of water scarcity is usually measured by the Water Stress Index (WSI)[29,65,66], which is defined as the ratio of annual water withdrawals (for domestic, industrial and agricultural uses) to the mean of available renewable water resources. The WSI indicator has been widely used to assess the status of water availability at the catchment or regional scales[67–69]. The regional WSIs within the intervals of 0–0.1, 0.1–0.2, 0.2–0.4, 0.4–0.8 and >0.8 represent respectively low, low to medium, medium to high, high, and extremely high levels of water stress[29]. Without the consideration of the effect of WWTPs, the urban water stress of city i was computed as follows:

$$\mathrm{WSI}_i = \frac{\mathrm{WU}_i}{\mathrm{RWR}_i} \quad (4)$$

where WU is the quantity of water distributed to end users, including water lost in transmission, which is equal to water withdrawals plus net physical water imports; and RWR denotes the total amount of renewable water resources (surface water and groundwater) available in a city[69]. In the absence of data on net physical water transfer at the city level, we calculated the local *RWR* by assuming that cities consume most of their required water from local sources.

To measure the alleviation effect attributable to WWTPs, we calculated the alternative water stress (WSI′) by incorporating two additional

factors in the equation: (1) the replenishment of treated wastewater to the surface renewable water in cities that has met the strict discharge standard of effluent (here this refers to China's I-B standard in the 'Discharge Standards of Pollutants for Municipal Wastewater Treatment Plants' (GB 18918–2002) (ref. 18), and (2) the reuse of water reclaimed from WWTPs in various urban activities including industrial use, municipal water and landscaping. Thus, WSI′ was used to measure the urban water stress of city *i* with the application of WWTPs:

$$\mathrm{WSI}_i' = \frac{\mathrm{WU}_i - \mathrm{RW}_i}{\mathrm{RWR}_i + \mathrm{RWW}_i} \quad (5)$$

where RW refers to the volume of reclaimed water reused in a city, and RWW refers to the volume of treated wastewater meeting the discharge standards, which thus can be treated as replenishment to the surface renewable water, for example, lakes, rivers and reservoirs.

ΔWSI refers to the relative variation between WSI and WSI′, which is calculated to represent the extent to which WWTPs have alleviated urban water stress and enhanced local water availability of city *i* (equation (6)). To further identify the importance of such variation to alleviating water stress, we further identify cities that have made 'major progress in WSI decrease'. In these cities, water stress has been downgraded from one status to another (for example, from an 'extreme water stress' to a 'high water stress' level).

$$\Delta\mathrm{WSI}_i = \frac{\mathrm{WSI}_i' - \mathrm{WSI}_i}{\mathrm{WSI}_i} \times 100\% \quad (6)$$

Scenario analysis

We simulated the future alleviation of water stress and life-cycle GHG emissions related to WWTPs by 2030 based on a set of scenarios that reflect the stepwise technological developments in their life-cycle stages including wastewater treatment, water reuse and sludge disposal. Towards this end, we first projected the future scales of urban wastewater treated in all the studied cities using an econometric model (Supplementary Note 3).

By combining treatment scale projection and technological transition, we set five policy scenarios to simulate alleviation on water stress and life-cycle GHG emissions induced by WWTPs from 2015 (the benchmark year) to 2030 (Supplementary Table 8). In the BAU scenario, the volume of reclaimed water increased in proportion to the scale of wastewater treatment, and the distribution proportion among technologies remained the same as in 2015. In the other four scenarios, we applied stepwise improvement strategies building on the BAU scenario. In the WRE scenario, the volume of reclaimed water increased from 2.5 Gt in 2015 to 8.2 Gt in 2020 according to the '13th Five-Year National Urban Sewage Treatment and Recycling Facilities Construction Plan'[19], which then was assumed to increase to 13.3 Gt simply in proportion to the scale of wastewater treatment in 2030.

We considered the currently available technologies of wastewater treatment, that is, PC treatment, ANA treatment, OD, A^2O/AO treatment, SBRs and MBRs; sludge disposal, that is, land use, landfill, incineration and reuse as building material; and water recycling and reuse that have the lowest GHG emissions intensities in each region as the most realistic and accessible low-carbon technologies by 2030 (a detailed description is presented in Supplementary Table 9). We then simulated the mitigation of the life-cycle GHG emissions of urban WWTPs in subsequent scenarios, with the assumption that a wider application of these technologies by 2030. Furthermore, we assume that, by 2020, 50% of volume of wastewater treated, scale of sludge disposal, or amount of water reuse will employ advanced low-carbon technologies (the remaining will be treated by legacy technologies). This proportion is expected to gradually rise to 75% by 2025 and 100% by 2030.

In the integrated scenario of WRE with low-carbon wastewater treatment technology (WRE + LCW), the technological mix for wastewater treatment have seen improvements using more advanced technologies that have lower carbon intensities, that is, a 7–37% reduction in wastewater treatment over the 2015–2030 period depending on the number of WWTPs that adopt these technologies across cities. Based on the WRE + LCW scenario, the WRE + LCW + LCSAD scenario further includes low-carbon sludge disposal with anaerobic digesters, whereby as a result of CH_4 recovery and sludge reduction the GHG emissions intensity in landfills, land use, incineration and reuse as building materials could be reduced respectively by 71%, 31%, 32% and 86% (ref. 39). This amounts to a 44–59% reduction in emission intensity of sludge disposal over 2015–2030. Finally, in the WRE + LCW + LCSAD + LCR scenario, low-carbon water reuse technologies are assumed to be applied more widely in cities. Therefore, in this scenario, compared with 2015, the GHG emissions intensity of water reuse is expected to be reduced by 54% in 2025 and 62% by 2030. This integrated scenario is used to assess whether improvement in technologies for reusing water and other life-cycle stages could effectively lower the GHG emissions that are caused by the intensification of water reuse.

We performed an uncertainty analysis for the simulation results in our study (see detailed description and measurements in Supplementary Note 4 and Supplementary Table 4, respectively).

Data

The plant-level WWTP monitoring data are compiled from the China Environmental Statistics Database (CESD)[55], the official Chinese database for statistics in the field of environmental protection. The CESD covers all the major urban WWTPs operated in China during 2006–2015. In 2015, the CESD covers more than 6,000 urban WWTPs distributed across >300 cities in 30 mainland provincial regions (excluding Tibet). The key plant-level information recorded in this database includes geographic location; year of construction; volume of wastewater treatment and treatment technologies, that is, PC treatment, ANA treatment, OD, A^2O/AO treatment, SBRs and MBRs; volumes of discharge into various types of water bodies (for example, rivers, lakes, reservoirs and the sea); approaches and scales of reclaimed water reuse, that is, industrial use, municipal water and landscaping; electricity consumption; amount of sludge disposed using different approaches, that is, land use, landfill, incineration and reuse as building materials; concentrations of chemical oxygen demand; ammonia nitrogen; total phosphorus in influent and effluent; and annual operating load and expenditure.

For assessing the water–climate nexus related to wastewater treatment, we aggregated various data to the city level. Towards this end, we analysed the changes of WSI at the city level based on water resources and consumption data that are available for more than 300 Chinese cities. We then aggregated unit-level WWTP data to the city level, for example, the total volume sum of wastewater treated at each WWTP, and matched them with the urban water stress results (see detailed description in Supplementary Note 5). The electricity use intensity of wastewater treatment of a city was calculated by the total electricity consumed by all WWTPs within a city divided by the city's total volume of treated wastewater. The upstream life-cycle GHG emissions of WWTPs were modelled on the basis of the established W-MRIO model and the provincial-level greenhouse emission intensities account, whereby the indirect GHG emissions of cities were separated on the basis of their annual investment in WWTPs. The detailed process of the integration and analyses of data at different levels of this study are explained in Supplementary Fig. 18.

Data availability

Sources of data used to perform this study are provided in Methods and Supplementary Information. Any further data that support the model of this study are available from the corresponding authors upon request.

Code availability

The programming code for the hybrid life-cycle model is available from the corresponding authors on request.

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Author contributions

S.C. and B.L. designed the research; L.Z., S.C. and H.S. performed the research; S.C., L.Z., H.Y., A.K. and F.J. analysed data; S.C., B.L., A.K., B.C., Z.L. and J.C.C. wrote the paper; and B.C. and J.C.C. reviewed and edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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